# Team Science Statement: Team Early Signal

*PoultryRescue · NAFSI Hub Student Data Challenge, Fall 2026, Sens-D track*

Team Early Signal formed organically when five individuals interested in the NAFSI HUB Data Challenge connected to participate. As work progressed, our complementary backgrounds in data analytics, computer science, engineering, food science, and nursing became important strengths. This interdisciplinary expertise allowed the team to consider not only how to develop a functioning analytical tool but also how its outputs could inform real-world food-safety and food-security decisions.

The computational and engineering perspectives supported the integration and analysis of public datasets, predictive modeling, software development, and implementation of the decision-support framework. Food science expertise contributed an understanding of poultry processing, microbial controls, and the nutritional implications of food loss. The nursing and population-health perspective connected Salmonella exposure and prevention with food insecurity, access to nutritious food, and the needs of under-resourced populations.

## Team roles

**Blessing Ademola** (PhD student, Chemical and Biomedical Engineering, University of Missouri): project conceptualization and planning, data integration, and validation of the state forecast.

**Friederike Albrecht** (PhD student, Sinclair School of Nursing, University of Missouri): review of the 2024 FSIS regulatory framework, the product-specific decision tree, the initial Python framework in Kiro, and the food-security and health-equity framing.

**Javonte Carter** (MS student, Computer Science, Florida A&M University): computing and software perspective, design discussions, and review of the application from a user's perspective.

**Nafisa Habib Purba** (PhD student, Food and Hospitality Systems, University of Missouri): food-safety and nutrition review of the decision framework, review of the analytical workflow, and outreach to University of Missouri food science specialists.

**Muhammad Waqar Saleem** (PhD student, Civil and Environmental Engineering, University of Missouri): data integration, the published-category forward test, the decision engine, the web application and repository, and the Kiro work.

## How different disciplines changed the work

The project began as a state-level case forecast. When alert precision fell from 64.4% in 2024 to 16.7% in 2026, the team kept the forecast as context only and moved the decision to the plant: the QA or HACCP lead at a small or very small ground poultry plant deciding what to do with one lot. Nursing and food-security input made servings kept, protein preserved and servings directed to high-need counties core outputs alongside exposure. Regulatory and food-science review added the 2024 rule for raw breaded stuffed chicken at 1 CFU/g or more, and the Listeria and cold-chain conditions that apply before cooked product can be donated.

## Value of the added data

Archived FSIS category postings and the MPI directory made the plant-level test possible: category plus product predicted next-year positivity 4.3% better than product alone (Brier score, 95% CI 2.9 to 6.0). CDC/ATSDR SVI, the USDA Food Access Research Atlas, CDC PLACES food insecurity and IRS food organizations target donations to high-need counties, and USDA cooking yields and FoodData Central convert a lot into servings and protein. Our ablation showed that adding FSIS, weather and storm data did not improve the state forecast, and we report that result.

## Next partner or dataset

The tool would gain most from QA and HACCP staff at small ground poultry plants to test it prospectively, a regional food bank to confirm donation capacity and cold chain, and enumeration and serotype data to move from positivity toward exposure.

Through ongoing discussion and the exchange of disciplinary perspectives, the team developed a processing-plant decision-support application that integrates predictive surveillance, product-specific regulatory considerations, safe-processing alternatives, nutritional preservation, and equitable food distribution. Together, our work shows how an initially informal, interdisciplinary team can build an integrated, data-driven approach to interconnected food-safety and food-security challenges.